Chapter 12. Planar Graphs

Do Not Cross

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12.1 Euler's theorem for planar graphs

Motivation. Can three houses be joined to three wells by non-crossing roads? Drawing a graph without crossings creates a new parameter, the number of faces, and Euler's formula $V+F=E+2$ ties it to V and E. Counting edges around faces then rules out $K_{3,3}$ and K_5.

Definition 12.1 (Planar graph)

A graph is **planar** if it can be drawn in the plane with no two edges crossing. The edges of such a drawing cut the plane into regions, the **faces**; one of them, the outer face, is unbounded.

Theorem 12.2 (Euler's theorem on planar graphs)

A connected planar graph with V vertices, E edges and F faces satisfies $V+F=E+2$.

Proof. Induction on E. For $E=1$: either one edge ($V=2$, $F=1$) or one loop ($V=1$, $F=2$). Let $E \geq 2$.

Step 1 (an edge on a cycle). If removing some edge e leaves G' connected, then e lies on a cycle, so the two sides of e belong to different faces, which merge in G'. G' has V vertices, $E-1$ edges and $F-1$ faces, so $V+F-1=E-1+2$ by induction.

Step 2 (a tree). Otherwise G is a tree, so $V=E+1$ (Theorem 10.4) and $F=1$. ■

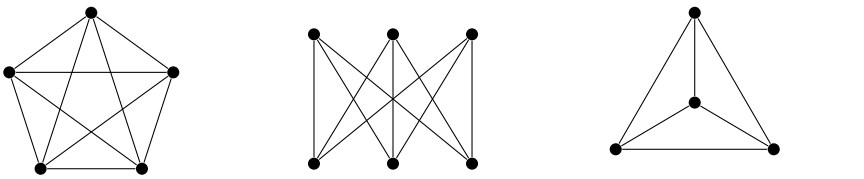
Example 12.3

$K_{3,3}$ is not planar, so the three houses, three wells problem has no solution.
Solution. With $V=6$, $E=9$, a planar drawing would have $F=5$ faces. $K_{3,3}$ is bipartite and simple, so it has no odd cycles and no 2-cycles, and every face is bounded by at least 4 edges. Each edge borders at most two faces, so $4F \leq 2E$, i.e. $20 \leq 18$, a contradiction.

$K_{3,3} \subseteq K_6$, so K_6 is not planar either. K_4 is planar (draw a square with one diagonal inside and the other outside).

Example 12.4

K_5 is not planar.
Solution. With $V=5$, $E=10$, a drawing would have $F=7$ faces, each bounded by at least 3 edges (the graph is simple). Then $3F \leq 2E$, i.e. $21 \leq 20$, a contradiction.



K_5 $K_{3,3}$ K_4, drawn without crossings

Figure. *The two minimal non-planar graphs and a crossing-free drawing of K_4.*

Deleting a vertex of degree 2 and merging its two edges, or subdividing an edge by a new vertex, does not change planarity. Graphs related by a sequence of these operations are **edge-equivalent** (homeomorphic).

Theorem 12.5 (Kuratowski)

A graph is not planar iff it contains a subgraph that is edge-equivalent to K_5 or $K_{3,3}$.

Proof. [Supplement] The book proves only the "if" direction: a subgraph edge-equivalent to K_5 or $K_{3,3}$ is non-planar by Examples 12.3 and 12.4, and a graph with a non-planar subgraph is non-planar. We prove the converse following Thomassen's argument. Loops and multiple edges do not affect planarity, so all graphs are simple. A subgraph edge-equivalent to X is called a TX (its *branch vertices* correspond to the vertices of X). X is a *minor* of G if there are disjoint vertex sets $V_h \subseteq V(G)$ $(h \in V(X))$, each inducing a connected subgraph, with an edge of G between V_h and $V_{h'}$ whenever $hh' \in E(X)$. A TX in G makes X a minor of G, and a minor of a minor is a minor. We use two standard facts about plane drawings: every face of a 2-connected plane graph is bounded by a cycle, and any face of a drawing can be made the outer face. ■

Lemma A (minors versus subdivisions)

If $K_{3,3}$ is a minor of G, then G contains a $TK_{3,3}$. If K_5 is a minor of G, then G contains a TK_5 or a $TK_{3,3}$.

Proof. [Supplement] Fix branch sets V_h and, for every edge hh' of X, one edge of G joining V_h and $V_{h'}$; its end in V_h is an *attachment point* of V_h. In each V_h take a minimal tree T_h containing all attachment points; its leaves are attachment points.

Step 1 ($X = K_{3,3}$). Each T_h has at most 3 leaves, so it is a path or a subdivided star with 3 rays. Choosing as branch vertex the centre of the star (or an end of the path, with the remaining attachment point on it), the trees together with the chosen edges form a $TK_{3,3}$.

Step 2 ($X = K_5$). Each T_h has at most 4 leaves. If every T_h has a vertex from which disjoint paths reach all its attachment points, these vertices are the branch vertices of a TK_5. Otherwise some T_h is a tree with 4 leaves and two vertices u, v of degree 3: the leaves joined to h_1, h_2 hang at u and those joined to h_3, h_4 hang at v. Then $\{u, h_3, h_4\}$ and $\{v, h_1, h_2\}$ are the two classes of a $K_{3,3}$-minor: u reaches h_1, h_2 and v through T_h, v reaches h_3, h_4, and $h_i h_j$ $(i \le 2 < j)$ are edges of K_5. In this minor every vertex has degree 3, and Step 1 applied to it gives a $TK_{3,3}$. ■

Lemma B (contractible edges)

Every 3-connected graph G with more than 4 vertices has an edge xy such that G/xy (contract xy to one vertex) is 3-connected.

Proof. [Supplement] Suppose not. For each edge xy, G/xy has a separating set of at most 2 vertices; as G is 3-connected it must contain the contracted vertex, so $\{x, y, z\}$ separates G for some z. Choose xy and z so that $G - \{x, y, z\}$ has a component C with as few vertices as possible.

Step 1. z has a neighbour $v \in C$: otherwise $\{x, y\}$ would separate C from the rest, contradicting 3-connectivity.

Step 2. By assumption $\{z, v, w\}$ separates G for some w. Since xy is an edge, $\{x, y\} \setminus \{w\}$ lies in one component of $G - \{z, v, w\}$, so there is another component D avoiding x and y. D contains a neighbour of v (Step 1's argument for v), which is not x, y, z, hence lies in C. As D avoids x, y, z it is connected in $G - \{x, y, z\}$, so $D \subseteq C$, and $v \notin D$ gives $|D| < |C|$. This contradicts the choice of C. ■

Lemma C (3-connected case)

Every 3-connected graph with no K_5- or $K_{3,3}$-minor is planar.

Proof. [Supplement] Induction on the number of vertices; with 4 vertices $G = K_4$, which is planar. Otherwise take xy as in Lemma B. G/xy is 3-connected and has no K_5- or $K_{3,3}$-minor, so by induction it has a plane drawing. Let w be the contracted vertex. $G/xy - w$ is 2-connected, so the face of $G/xy - w$ containing w is bounded by a cycle C, and all neighbours of x and y other than x, y lie on C. Since G is 3-connected, x has neighbours $x_1, \dots, x_k$ ($k \geq 2$) on C in cyclic order, which cut C into arcs P_i from x_i to x_{i+1}; y has at least 2 neighbours on C.

Step 1. If all neighbours of y on C lie on one arc P_i (endpoints included), redraw: put x at w with its edges to the x_j, and put y inside the region bounded by xx_i, P_i and $x_{i+1}x$, joined to x and to its neighbours on P_i. This is a plane drawing of G.

Step 2. Otherwise either y has three neighbours that are also neighbours of x, or y has neighbours y_1, y_2 such that x_a, y_1, x_b, y_2 occur in this cyclic order on C for some neighbours x_a, x_b of x. (Indeed, if some neighbour y_1 of y is inside an arc P_i, a neighbour outside P_i alternates with x_i, x_{i+1}; if all neighbours of y are among the x_j and there are only two, they are not consecutive, so other x-neighbours separate them.) In the first case x, y and the three common neighbours, joined by arcs of C, form a TK_5. In the second case the classes $\{x, y_1, y_2\}$ and $\{y, x_a, x_b\}$, joined by the edges at x and y and by the four arcs of C between consecutive vertices of x_a, y_1, x_b, y_2, form a $TK_{3,3}$. Either way G would have a K_5- or $K_{3,3}$-minor, which is excluded, so Step 1 applies. ■

Lemma D (reduction to the 3-connected case)

Let G have at least 4 vertices and be edge-maximal with no TK_5 and no $TK_{3,3}$ (adding any edge creates one). Then G is 3-connected.

Proof. [Supplement] Induction on $|V(G)|$; with 4 vertices $G = K_4$. Suppose $|V(G)| \geq 5$ and G has a separating set S with $|S| \leq 2$; write $G = G_1 \cup G_2$ with $G_1 \cap G_2 = S$, both $G_i - S$ nonempty.

Step 1 (S is complete). Take S minimal; then every $s \in S$ has a neighbour in every component of $G - S$. Suppose $S = \{x, y\}$ with $xy \notin E(G)$. Then $G + xy$ contains a TX ($X = K_5$ or $K_{3,3}$) using xy. Since X stays connected after deleting any two vertices, all branch vertices outside S lie on one side, say in G_1. Then TX meets $G_2 - S$ in at most one path through S, and xy can be replaced by an x–y path through a component of $G_2 - S$. This gives a TX in G, a contradiction.

Step 2 (the parts). By the same rerouting, each G_i is edge-maximal without TK_5 and $TK_{3,3}$: a TX in $G_i + e$ would give one in $G + e$ and conversely, since paths through the other side can be replaced by the edge xy. By induction (if $|V(G_i)| \geq 4$) and Lemmas A and C, or trivially (if $|V(G_i)| \leq 3$), each G_i is planar.

Step 3 (gluing). Draw G_2 with the edge xy (or the vertex of S) on its outer face, and place it inside a face of G_1 incident with S. This is a plane drawing of G in which some face contains a vertex $a \in G_1 - S$ and a vertex $b \in G_2 - S$ on its boundary, so $G + ab$ is still planar. A planar graph contains no TK_5 or $TK_{3,3}$, so G is not edge-maximal. This contradiction shows G is 3-connected. ■

Proof of Theorem 12.5, "only if". [Supplement] Let G contain no subgraph edge-equivalent to K_5 or $K_{3,3}$. Add edges while this property is kept, to obtain an edge-maximal $G' \supseteq G$ on the same vertices. If G' has at most 3 vertices it is planar. Otherwise G' is 3-connected by Lemma D, and by Lemma A it has no K_5- or $K_{3,3}$-minor, so G' is planar by Lemma C. Hence $G \subseteq G'$ is planar. ■

12.2 Polyhedra

Motivation. Regular polygons exist with any number of sides, but there are only five regular polyhedra. The reason is purely combinatorial: Euler's formula, together with "every face has ≥ 3 edges and every vertex has degree ≥ 3", leaves room for only five pairs (d, ℓ).

For a convex polyhedron (identified with its 1-skeleton) every face has at least 3 edges and every vertex lies on at least 3 edges. A polyhedron is **regular** (combinatorially) if every face is an ℓ-cycle and every vertex has degree d.

Theorem 12.6

A convex polyhedron with V vertices, F faces and E edges satisfies $V + F = E + 2$.

Proof (angle counting). Project P onto a plane p perpendicular to no face. Every face projects to a convex polygon, and the outer boundary B of the image is a convex v_1-gon; let v_2 vertices project inside it, so $v_1 + v_2 = V$. Each point inside B is covered by exactly one "front" and one "back" face. Let S be the sum of all angles of all projected faces.

Step 1 (by vertices). At a vertex projecting inside B, the image faces meeting there fill a full neighbourhood, so their angles sum to $360°$; this gives $360v_2$. At a vertex on B, the front faces and the back faces each fill the interior angle of B, so the boundary vertices contribute twice the angle sum $(v_1 - 2)180°$ of B. Hence

$$S = (v_1 - 2)360 + 360v_2 = (V - 2)360. \tag{12.1}$$

Step 2 (by faces). Let $f_1, \dots, f_F$ be the numbers of edges of the faces. Each edge lies on two faces, so

$$\sum_{i=1}^{F} f_i = 2E, \tag{12.2}$$

and a convex k-gon has angle sum $(k - 2)180°$, so

$$S = \sum_{i=1}^{F} (f_i - 2)180 = 180 \sum_{i=1}^{F} f_i - 360F = 360(E - F). \tag{12.3}$$

Comparing (12.1) and (12.3) gives $V - 2 = E - F$. ∎

Corollary 12.7

In a convex polyhedron, $3F \le 2E$.

Proof. $f_i \ge 3$ in (12.2). ∎

Proposition 12.8

In a convex polyhedron, $3V \le 2E$.

Proof. Let $c_1, \dots, c_V$ be the vertex degrees. Each edge has two ends, so

$$\sum_{i=1}^{V} c_i = 2E, \tag{12.4}$$

and $c_i \ge 3$. ∎

Lemma 12.9

In a convex polyhedron, $E \le 3V - 6$ and $E \le 3F - 6$.

Proof. By Corollary 12.7, $E + 2 = V + F \le V + \frac{2E}{3}$, so $\frac{E}{3} \le V - 2$. By Proposition 12.8, $E + 2 \le F + \frac{2E}{3}$, so $\frac{E}{3} \le F - 2$. ∎

Remark

Remark[$E \le 3V - 6$ for every simple planar graph [Supplement]] Let G be simple and planar with $V \ge 3$; we may assume it connected (adding edges between components keeps it planar). If G is a tree, $E = V - 1 \le 3V - 6$. Otherwise every face is bounded by a closed walk of length at least 3 (no loops or multiple edges) and each edge borders at most two faces (counted twice if it borders one face on both sides), so $3F \le 2E$. Euler's formula then gives $E \le 3V - 6$ as above. Consequently every simple planar graph has a vertex of degree at most 5: otherwise $2E \ge 6V$.

Lemma 12.10

Every convex polyhedron has a face with at most five edges.

Proof. By (12.2) and Lemma 12.9,

$$\sum_{i=1}^{F} f_i = 2E \le 6F - 12 < 6F, \tag{12.5}$$

so not all $f_i \ge 6$. ■

Lemma 12.11

Every convex polyhedron has a vertex of degree at most five.

Proof. By (12.4) and Lemma 12.9,

$$\sum_{i=1}^{V} c_i = 2E \le 6V - 12 < 6V. \tag{12.6}$$

■

The symmetry between V and F has a structural reason. Given a planar G, put a vertex inside each face and, for every edge e of G, join the vertices of the two faces on either side of e by an edge crossing e. The result G^* is planar with F vertices, E edges and V faces.

Definition 12.12 (Dual graph)

G^* is the **dual graph** of the plane graph G.

A statement about (V, E) of a polyhedron P is a statement about (F, E) of P^*. The dual of a regular polyhedron of type (d, ℓ) is regular of type (ℓ, d).

Classification. In a regular polyhedron, Lemmas 12.10 and 12.11 give $d, \ell \in \{3, 4, 5\}$, and by (12.2)

$$E = \frac{F\ell}{2}. \tag{12.7}$$

(A) $d = 3$: $3V = 2E$, so Euler gives $3F = E + 6$, and with (12.7), $(6 - \ell)F = 12$. Then $\ell = 3, 4, 5$ give $(V, E, F) = (4, 6, 4)$ (tetrahedron), $(8, 12, 6)$ (cube), $(20, 30, 12)$ (dodecahedron).

(B) $d = 4$: $4V = 2E$, so $E = 2F - 4$ and $(4 - \ell)F = 8$; only $\ell = 3$ works: $(6, 12, 8)$, the octahedron (dual of the cube).

(C) $d = 5$: $5V = 2E$, so $3E = 5F - 10$ and $(10 - 3\ell)F = 20$; only $\ell = 3$ works: $(12, 30, 20)$, the icosahedron (dual of the dodecahedron).

Theorem 12.13

There are exactly five regular polyhedra: the tetrahedron, the cube, the dodecahedron, the octahedron and the icosahedron.

Proof. The case analysis (A)–(C) leaves exactly these five parameter sets, and each is realised by the named solid. ■

	d	ℓ	V	E	F
Tetrahedron	3	3	4	6	4
Cube	3	4	8	12	6
Octahedron	4	3	6	12	8
Dodecahedron	3	5	20	30	12
Icosahedron	5	3	12	30	20

12.3 Colouring maps

Motivation. Colouring the countries of a map so that neighbours differ is, by duality, properly colouring the vertices of a planar graph. A vertex of degree ≤ 5 gives six colours at once; Kempe's chain-swapping trick brings this down to five. Four colours suffice too, but every known proof uses a computer.

Faces sharing an edge must get different colours; this is the same as properly colouring the vertices of G^*. K_4 is planar, so at least four colours are needed.

Proposition 12.14

Every planar graph can be properly vertex-coloured with six colours.

Proof. Induction on V; loops are excluded and multiple edges do not matter, so let G be simple. By the Remark after Lemma 12.9, G has a vertex A of degree at most 5. Colour $G - A$ with six colours by induction, and give A a colour not used by its at most five neighbours. ■

Theorem 12.15 (Five colour theorem)

Every planar graph can be properly vertex-coloured with five colours.

Proof. Induction on V for simple G. Take A of degree ≤ 5 and colour $G' = G - A$ properly with five colours. If the neighbours of A use at most four colours, colour A with the fifth. Otherwise A has neighbours $y_1, \dots, y_5$ in clockwise order, coloured $1, \dots, 5$.

Step 1 (Kempe chains). Let H_{13} be the subgraph of G' induced by the vertices coloured 1 or 3, and K the component of H_{13} containing y_1. Swapping colours 1 and 3 on K keeps the colouring proper. If $y_3 \notin K$, after the swap y_1 and y_3 both have colour 3, colour 1 is free at A, and we are done.

Step 2 (both chains cannot exist). So assume $y_3 \in K$: there is a path $y_1 \to y_3$ coloured alternately $1, 3$. Together with A it forms a cycle C through A that separates y_2 from y_4 in the plane (the neighbours occur around A in the order y_1, y_2, y_3, y_4). By the same argument with colours $2, 4$, we are done unless there is a path $y_2 \to y_4$ coloured $2, 4$. This path would have to cross C at a vertex, but its vertices are coloured $2, 4$ and those of $C - A$ are coloured $1, 3$. So this case cannot occur. ■

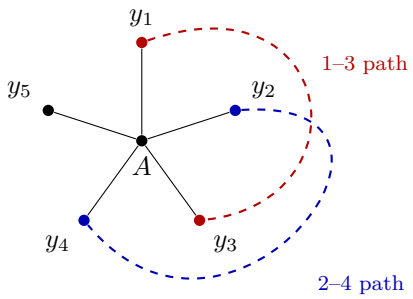


Figure. *Kempe chains in the five colour theorem: a 1–3 path from y_1 to y_3 and a 2–4 path from y_2 to y_4 would have to cross at a vertex.*

Remark

Remark[The four colour theorem] Four colours always suffice for maps with contiguous countries. Appel and Haken proved this in 1976 by a computer check of 1936 configurations, later reduced to 633 by Robertson, Sanders, Seymour and Thomas (1996). A machine-verified proof was given by Gonthier and Werner (2005).

12.4 Selected exercises

Motivation. In the book exercises below, Euler's formula is combined with two double counts (edges around faces, edges at vertices); the face lengths and degrees then satisfy a single linear relation that decides everything.

Throughout, the *length* of a face is the number of edges on its boundary, an edge counted twice if the face lies on both of its sides. Every edge has two sides, so the face lengths add up to $2E$; the degrees add up to $2E$ as well.

Exercise 12.15 (Polyhedra without triangles)

Let P be a convex polyhedron with no triangular faces. Prove that $E \le 2V - 4$.

Solution.

Step 1 (faces). Every face has at least 4 edges, so $4F \le 2E$, i.e. $F \le \frac{E}{2}$.

Step 2 (Euler). By Theorem 12.6, $E + 2 = V + F \le V + \frac{E}{2}$, so $\frac{E}{2} \le V - 2$ and $E \le 2V - 4$.

Equality holds iff all faces are quadrilaterals, e.g. for the cube ($V = 8$, $E = 12$). ■

Exercise 12.19 (Trivalent (a, b)-faced polyhedra)

Let P be a convex polyhedron whose faces are all a-gons or b-gons and in which every vertex lies on exactly three edges. Let p_a, p_b be the numbers of a-gonal and b-gonal faces and n the number of vertices.
(a) Express the number E of edges in two ways.
(b) Prove that $p_a(6 - a) + p_b(6 - b) = 12$.

Solution.

(a) Counting edges around faces, $2E = a\,p_a + b\,p_b$. Counting edges at vertices, $2E = 3n$.

(b) By Theorem 12.6, $n + p_a + p_b = E + 2$. Multiply by 6 and use $6n = 4E$ from (a):

$$4E + 6p_a + 6p_b = 6E + 12, \qquad \text{so} \qquad 6p_a + 6p_b - 2E = 12.$$

Substituting $2E = a\,p_a + b\,p_b$ gives $p_a(6-a) + p_b(6-b) = 12$.

For example, a trivalent polyhedron with pentagonal and hexagonal faces only (a fullerene, such as the soccer ball) has $p_5 \cdot 1 + p_6 \cdot 0 = 12$: exactly 12 pentagons, whatever the number of hexagons. ■

Exercise 12.22 (Octahedrites)

Let G be a connected planar graph in which each face has length 2, 3 or 4, with p_2, p_3, p_4 faces of these lengths. Assume that every vertex has degree 4 and that $p_2 + p_3 = 8$. Prove that $p_2 = 0$ and $p_3 = 8$.

Solution.

Step 1 (two counts). Counting at the vertices, $4V = 2E$, so $V = \frac{E}{2}$. Counting around the faces, $2E = 2p_2 + 3p_3 + 4p_4$, and $F = p_2 + p_3 + p_4$.

Step 2 (Euler). By Theorem 12.2, $V + F = E + 2$, i.e. $F = \frac{E}{2} + 2$. Multiplying by 4,

$$4p_2 + 4p_3 + 4p_4 = 2E + 8 = 2p_2 + 3p_3 + 4p_4 + 8, \qquad \text{so} \qquad 2p_2 + p_3 = 8.$$

Step 3 (conclusion). Subtracting $p_2 + p_3 = 8$ gives $p_2 = 0$, and then $p_3 = 8$. ■